

CV

Apurva Nakade

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Experience

Position	Institution	Year
Assistant Teaching Professor	Johns Hopkins University	2026-Present
Senior Lecturer	Johns Hopkins University	2023-2026
Postdoctoral Lecturer	Northwestern University	2021-2023
Postdoctoral Fellow	University of Western Ontario	2019-2021
Ph.D. in Mathematics	Johns Hopkins University	2019
Academic Coordinator	Canada/USA Mathcamp	2018
Mentor	Canada/USA Mathcamp	2017-2020
M.Sc. in Mathematics	Chennai Mathematical Institute	2013
Exotic Derivatives Trader	Nomura Capital India Pvt Ltd	2010
B.Tech. in Computer Science & Engineering	IIT Kanpur	2010

Teaching

Teaching Evaluations

Instructor, JHU AMS

Spring 2026

- Intro to Computational Math, Spring 2026
- Discrete Math, Spring 2026
- Monte Carlo Methods, Spring 2026

Fall 2025

- Monte Carlo Methods, Fall 2025
- Discrete Math, Fall 2025

Summer 2025

- Mathematical Foundations of AI, Summer 2025
- Intermediate Probability and Statistics, Summer 2025
- Honors Algebra II, Summer 2025

Spring 2025

- Discrete Math, Spring 2025
- Monte Carlo Methods, Spring 2025

- Honors Algebra II, Spring 2025
- Intermediate Probability and Statistics, Spring 2025

Fall 2024

- Discrete Math, Fall 2024
- Monte Carlo Methods, Fall 2024
- Honors Algebra II, Fall 2024

Summer 2024

- Exploring Engineering Innovation, Summer 2024

Spring 2024

- Graph theory, Spring 2024
- Linear Algebra and Differential Equations, Spring 2024
- Honors Algebra II, Spring 2024

Fall 2023

- Discrete Math, Fall 2023

Instructor, Northwestern

- Elementary Differential Equations, Spring 2023
- MENU Linear Algebra and Multivariable Calculus, (Coordinator) 2022-23
- Introduction to Optimization, Winter, Spring 2022
- Single Variable Calculus, Fall 2021
- MENU Linear Algebra and Multivariable Calculus, 2021-22
- Foundations of Mathematics, Winter 2023
- Elementary Differential Equations, Spring 2023

Instructor, UWO

- Algebraic Topology (graduate level), Winter 2021
- Topology Bootcamp, Fall 2020
- Discrete Structures for Engineering, Fall 2020
- Calculus II for Mathematical and Physical Sciences, Winter 2020
- Calculus I for Mathematical and Physical Sciences, Fall 2019
- Topics in Category Theory, Fall 2019

Instructor, JHU

- Honors Single Variable Calculus, Fall 2018, 2017
- Symmetries & Polynomials, Intersession 2018
- Hitchhikers Guide to Algebraic Topology, Intersession 2017
- Differential Equations with Applications, Summer 2017, 2015
- Online Linear Algebra, Summer 2014

Academic Co-coordinator, Canada/USA Mathcamp

- Planned the five week academic schedule

- Part of the mentor hiring committee
- Invited visiting speakers

Mentor/Staff, Canada/USA Mathcamp

- Designed and taught a variety of undergraduate-level courses
- Was residential and academic advisor at camp
- Part of the mentor hiring committee - Taught a course at Mathcamp 2020
- Contributed to Lean's surreal numbers math library
- Contributed to Lean's convex optimization math library

Professional Development

Faculty Forward Fellowship, JHU, 2025

Introduction to Education Research Workshop, JHU, 2023

MAA Section NExT Fellow, 2023 cohort, 2023-25

Mathematical Association of America Fellow, 2020-Present

MAA Project NExT Fellow, Brown'20 cohort, 2020

Teaching Academy Certification, JHU, 2019

Participated in several workshops by the Center of Teaching & Learning at UWO, 2019-2020

Participated in the MSRI Critical Issues in Mathematics Education Workshop, 2022

Participated in the MAA Modeling Inspiration for Differential Equations Workshop, 2022

Participated in online Mastery Grading Workshop, 2019

Attended the Science of Learning Symposium, JHU, 2014-2018

Projects

Course Design: Discrete Mathematics, JHU

- Modernized the course using evidence-driven pedagogical techniques
- Adapted an online textbook to better match current student mindsets
- Introduced homework resubmissions to reinforce student learning
- Introduced online assessments to improve feedback
- Restructured exams to better gauge student learning

Open Educational Resources Textbook for Linear Algebra, NU

- Added WeBWork problems to a Linear Algebra PreTeXt OER textbook

Course Design: Introduction to Optimization, UWO

- Restructured the course to include applications and modeling
- Created course notes in RMarkdown
- Created Excel worksheet assignments for modeling scenarios

Course Design: Discrete Structures for Engineering, UWO

- Adapted the course for online asynchronous delivery
- Helped code (in a team) hundreds of problems on WeBWork
- Made short weekly video lectures to support the course text
- Maintained an active discussion forum on Piazza
- Gave a talk about the course design at a conference on E-Assessment in Mathematical Sciences

Course Design: Honors Single Variable Course (IBL), JHU

- Designed and taught a semester long course structured in a flipped classroom format for two semesters

Course Design: Intersession courses, JHU

- Designed and taught a 2-week course titled Symmetries & Polynomials introducing Galois theory to non-math majors
- Designed and taught a 2-week course Hitchhiker's Guide to Algebraic Topology introducing algebraic topology to non-math majors

Course Design: Canada/USA Mathcamp courses

- Designed and taught several week-long math courses to advanced highschool students
- Course topics: Theorem proving in Lean, Visualizing 3-manifolds, Riemann surfaces, Crash Course on Linear Algebra, Covering Spaces, Cohomology etc.

Papers

String structures, 2-group bundles, and a categorification of the Freed-Quinn line bundle, Daniel Berwick-Evans, Emily Cliff, Laura Murray, Apurva Nakade, Emma Phillips, Quantum Symmetries, Contemporary Mathematics, vol. 813, Amer. Math. Soc., Providence, RI, 2025, pp. 257-301.

Manifold Calculus and the h-principle, The Journal of Homotopy and Related Structures, 2019

Effect of increasing the energy gap between the two lowest energy states on the mixing time of the Metropolis algorithm (with Somenath Biswas), Information Processing Letters, IPL4801 (2012.08.012), 2012

Grants & Awards

Open Educational Resources Faculty Grant (joint with Dhruv Azad), JHU, 2026

- \$5000 to develop interactive visualization apps for computational math courses

Open Educational Resources Faculty Grant (joint with Aaron Greicius), NU, 2022

William Kelso Morrill Award for Excellence in Mathematics, JHU, 2019

Finalist for the KSAS Excellence in Teaching Awards, JHU, 2019

Prof. Joel Dean Award for Excellence in Teaching in Mathematics, JHU, 2016

AMS Graduate Student Travel Grant, 2019

Service

Member of DS Masters admissions committee in JHU AMS department, 2025-2026

Helped connect undergraduate students with research mentors in JHU AMS department, 2024-2025

Judge for poster presentations at JHU Matrix Conference, 2025

Judge for poster presentations at MAA MD-DC-VA Section Meeting, Fall 2024

Supplementary Instructor for the Causeway Postbaccalaureate Program, NU, 2022

Directed Reading Program, UWO, JHU, 2017-2021

- Started DRP at UWO in Fall 2019
- Organizer and mentor for DRP at JHU and UWO Professional Development